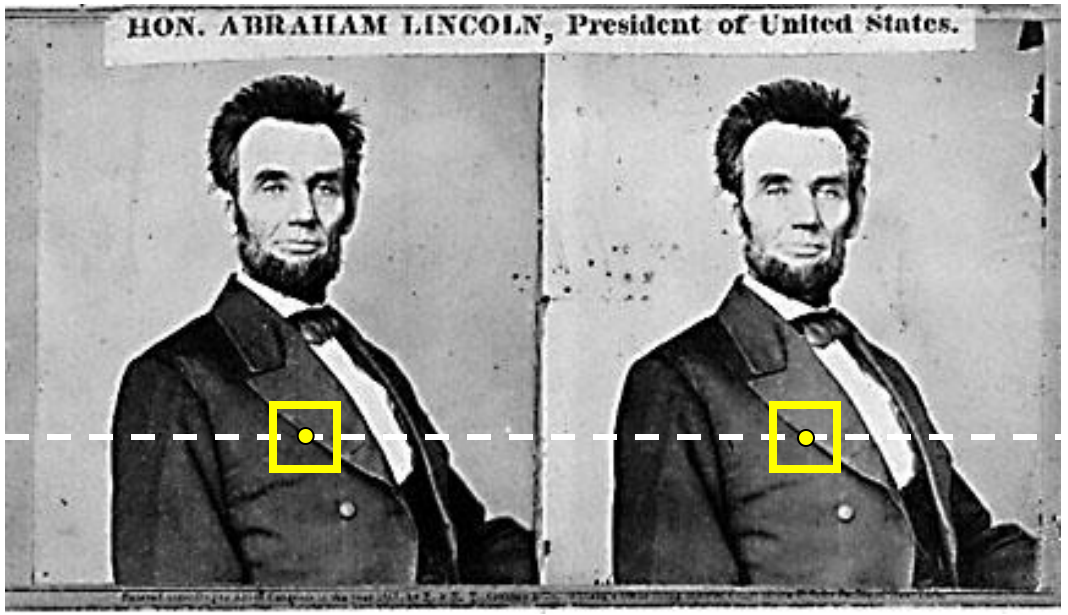
Your basic stereo matching algorithm



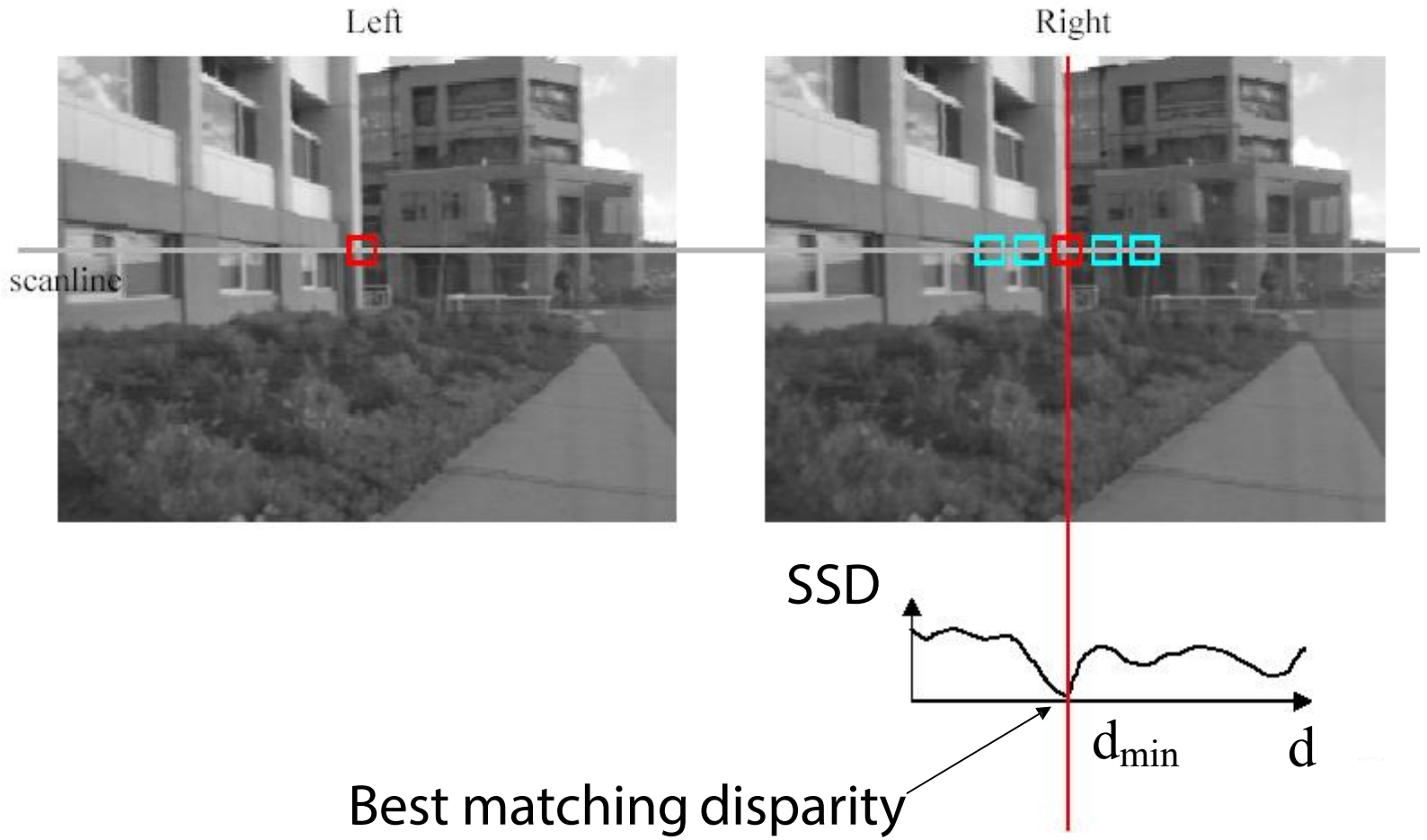
For each epipolar line

For each pixel in the left image

- compare with every pixel on same epipolar line in right image
- pick pixel with minimum match cost

Improvement: match ***windows***

Stereo matching based on SSD

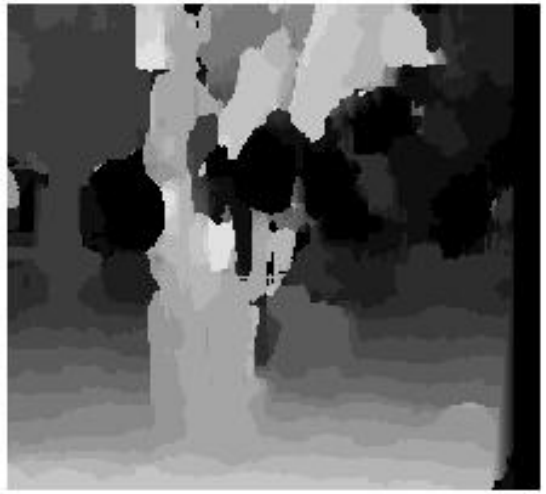


Window size





W = 3



W = 20

Effect of window size

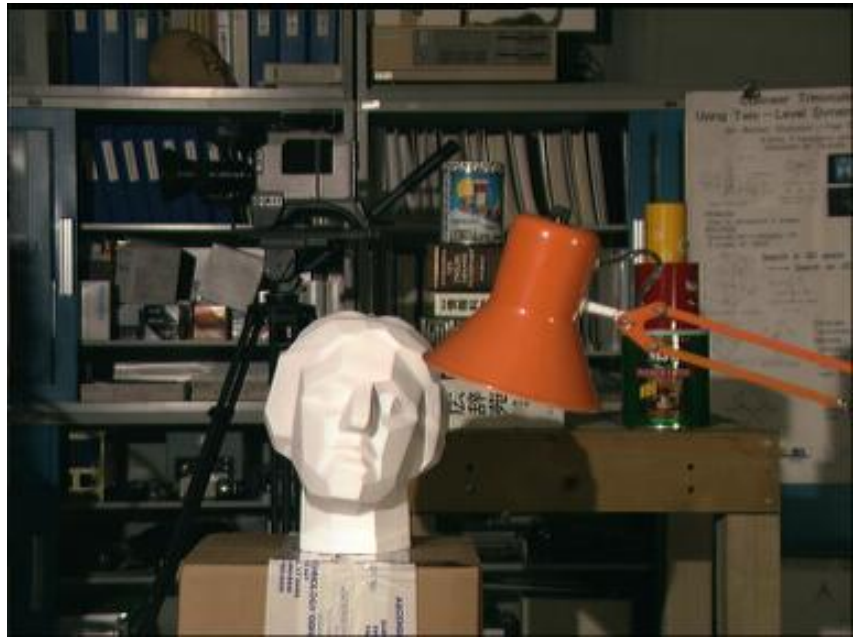
- Smaller window
  - \+ more detail
  - \- more noise
- Larger window
  - \+ less noise
  - \- less detail

Better results with *adaptive window*

- T. Kanade and M. Okutomi, *A Stereo Matching Algorithm with an Adaptive Window: Theory and Experiment*, ICRA 1991.
- D. Scharstein and R. Szeliski. Stereo matching with nonlinear diffusion. IJCV, July 1998

Stereo results

– Data from University of Tsukuba
– Similar results on other images without ground truth

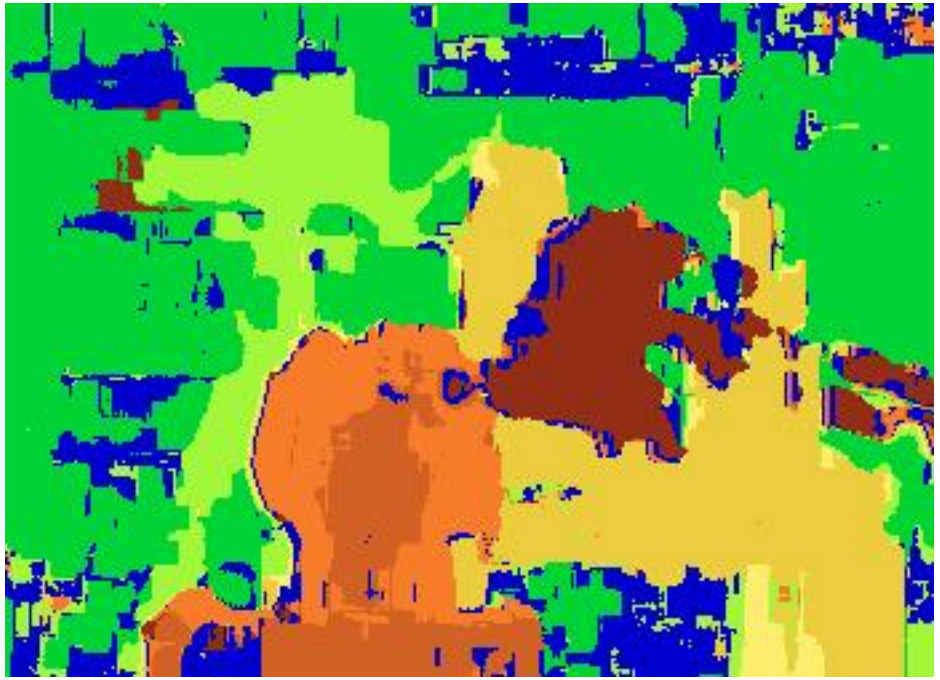


Scene



Ground truth

Results with window search



Window-based matching
(best window size)



Ground truth

Better methods exist...



Graph cuts-based method

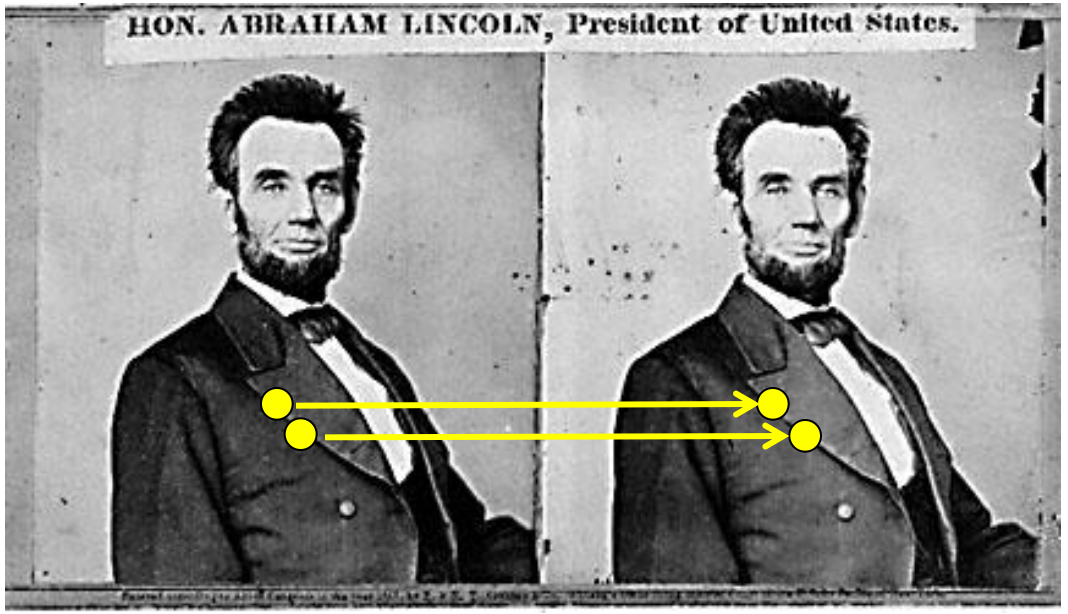


Ground truth

Boykov et al., Fast Approximate Energy Minimization via Graph Cuts, International Conference on Computer Vision 1999.

For the latest and greatest: http://www.middlebury.edu/stereo/

Stereo as energy minimization



- What defines a good stereo correspondence?
  1. Match quality
     - Want each pixel to find a good match in the other image
  2. Smoothness
     - If two pixels are adjacent, they should (usually) move about the same amount

Stereo as energy minimization

- Find disparity map *d* that minimizes an *energy function*

$$E(d)$$

- Simple pixel / window matching

$$E(d) = \sum_{(x,y) \in I} C(x, y, d(x, y))$$

$$C(x, y, d(x, y)) = \text{SSD distance between windows } I(x, y) \text{ and } J(x + d(x,y), y)$$